Record - Week 02

Topic: JavaScript ES6 and DOM

Name: _____ Roll No: _____ Date: _____

Aim

To manipulate the DOM using ES6 features and to fetch remote data asynchronously.

Software Requirements

Web browser, VS Code, Node.js (optional)

Procedure

1 Create index.html with an input, buttons and lists.

2 Write addTask() that builds li elements with createElement.

3 Attach click listeners for toggling and deleting tasks.

4 Write loadUsers() using async/await and fetch.

5 Handle errors with try/catch and test in the browser.

Program

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Week 02 - JavaScript ES6 &amp;amp; DOM</title>
</head>
<body>
  <h1>Task Manager</h1>
  <input id="taskInput" placeholder="Add a task">
  <button id="addBtn">Add</button>
  <ul id="taskList"></ul>

  <h2>Users (fetched)</h2>
  <button id="loadBtn">Load Users</button>
  <ul id="userList"></ul>

  <script src="app.js"></script>
</body>
</html>
```

app.js

```
// ---------- DOM manipulation ----------
const input = document.getElementById("taskInput");
const addBtn = document.getElementById("addBtn");
const taskList = document.getElementById("taskList");

const addTask = () => {
  const text = input.value.trim();
  if (!text) return;

  const li = document.createElement("li");
  li.textContent = text;
  li.addEventListener("click", () => li.classList.toggle("done"));

  const del = document.createElement("button");
  del.textContent = "x";
  del.addEventListener("click", (e) => {
    e.stopPropagation();
    li.remove();
  });

  li.appendChild(del);
  taskList.appendChild(li);
  input.value = "";
};

addBtn.addEventListener("click", addTask);

// ---------- Async / fetch ----------
const loadBtn = document.getElementById("loadBtn");
const userList = document.getElementById("userList");

async function loadUsers() {
  try {
    const res = await fetch("https://jsonplaceholder.typicode.com/users");
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const users = await res.json();
    userList.innerHTML = users
      .map(({ name, email }) => `<li>${name} - ${email}</li>`)
      .join("");
  } catch (err) {
    userList.innerHTML = `<li>Error: ${err.message}</li>`;
  }
}

loadBtn.addEventListener("click", loadUsers);
```

Output

Fig: dom manipulation output

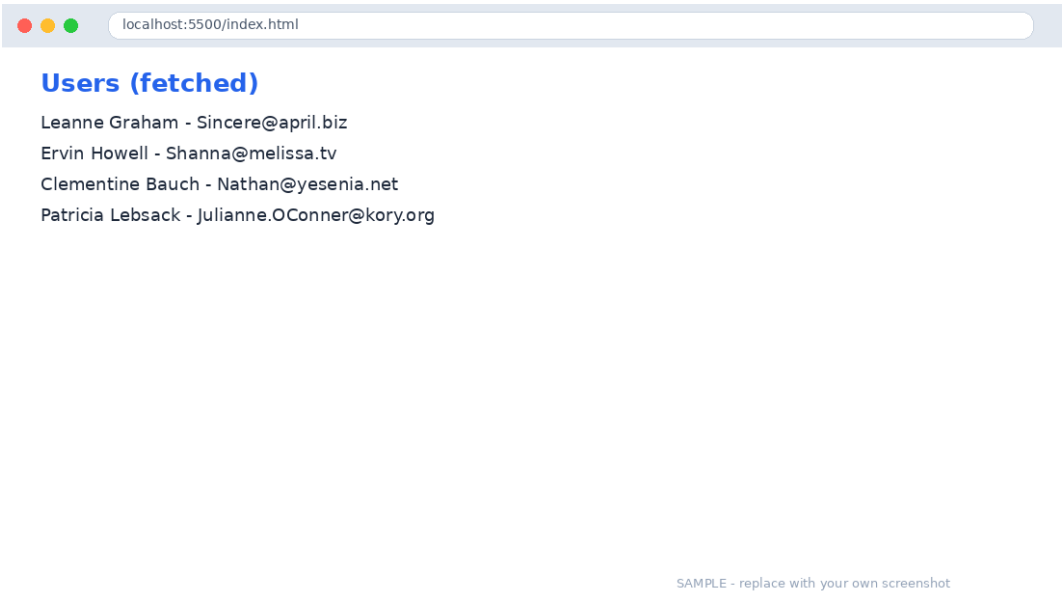

Fig: async fetch output

Result

The program was executed successfully and the expected output was obtained.